Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/21/2015
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	03/06/2014	**End Date:**	03/05/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Prof. Carlos Fernandez-Pello is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA PI is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AF01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. The objective of the proposed research program is to continue the experimental study of the flammability of wire materials in space exploration atmospheres and associated computational/theoretical tools to aid interpretation of test results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in space allows us to accurately control the flow field and thus elucidate the importance of a critical Damkohler number (flow time /reaction time) on flame extinction. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Investigation and results have Earth benefits for terrestrial fire safety.
Task Progress:	1. Normal gravity and microgravity flammability limits of ETFE (ethylene tetrafluoroethylene) insulated wires exposed to external radiation. Experiments were conducted of the flammability characteristics of sample wires of a 0.50 mm copper core and 0.30 mm ETFE insulation thickness. The experiments consisted on the determination of the ignition and flame spread potential of the wires in oxygen concentrations ranging from 20% to 32% and external radiant fluxes from 0 to 25 kW/m2. The apparatus and supporting equipment is designed to fit in an experimental parabolic flight rack for microgravity experiments. The main component of the apparatus is a 60 mm in diameter and 250 mm long flow duct made of Pyrex. The ETFE insulated wire is placed at the centerline of the duct and it is fed via a spool system located at both ends of the tube. The combustion products are filtered with an air filter located downstream of the duct. A suction fan attached to the air filter is used to induce a recirculating airflow throughout flow duct. Honeycomb flow straighteners placed at both ends of the tube are used to ensure flow uniformity. A constant airflow speed of 0.12 m/s through the flow duct was used for all experiments. This flow velocity simulates typical airflow speeds found in spacecraft ventilation systems. Two 300 mm long halogen heaters with a peak wavelength emission of 1.2 microns are mounted directly on top and below the flow duct. The heaters are contained in an isolated volume that limits exposure to toxic and corrosive ETFE combustion products. The heaters are controlled using a transformer and provided a maximum radiant flux of 25 kW/m2 to the wire sample. A heat flux sensor is used to verify a uniform heat flux distribution along the wire surface. ETFE insulated copper wire samples (Sugita Densen Co., Ltd.) with a core diameter of 0.50 mm and an insulation thickness of 0.30 mm are used in all experiments. Ignition of the wire insulation is accomplished using a 0.5 mm thick Kanthal wire coil with a 8 mm diameter. The coil is concentric with the wire and is placed downstream end of the wire. A programmable logic controller is used to control a constant current power supply that energizes the igniter coil for 12 s with a maximum power of 92 W. In microgravity experiments, the igniter was energized 10 seconds before the beginning of the ug period to reduce the influence of the igniter during ug flame spread. A pressure sensor and an oxygen sensor are used to monitor pressure and oxygen concentration in the chamber. One type-T thermocouple is used to measure air temperature in the chamber. The gas inside the chamber consists of mixtures of oxygen and nitrogen. The desired oxygen concentration inside the chamber is set using oxygen partial pressures with a total chamber pressure of 100 kPa. Once a test starts the external radiant heaters are turned on and allowed to irradiate the sample wire. The igniter is turned on after a 30 s preheating time. Microgravity experiments are conducted in parabolic flights onboard a MU-300 aircraft operated by Diamond Air Service (DAS) in Nagoya, Japan. Each parabola provides about 20 s of 10-2 g. Chamber pressure, oxygen concentration, air temperature, and acceleration in the case of microgravity experiments, were recorded using a data acquisition system. Experiments consisted in determining whether opposed flame spread occurred along the wire insulation for a given set of ambient oxygen concentration and external radiant flux conditions. The oxygen concentration resulting in a change from No Propagation to Propagation was defined as the Limiting Oxygen Concentration (LOC). The LOC values as a function of the applied external radiant flux are used to produce 1g and µg LOC boundaries. In the absence of an external radiant flux the LOC of the ETFE wire was found to be 31.5%, which is in reasonable agreement with ETFE LOI of 30% reported in the literature The addition of an external radiant flux extended the LOC of ETFE below 31.5%. At the maximum radiant flux of the apparatus (q”ext=24.3 kW/m2) the lowest oxygen concentration at which flame spread was observed was at 26.3. It is seen that as the oxygen concentration is reduced, larger values of the external radiant flux are required to sustain flame propagation. The number of tests that could be performed in microgravity was small due to the limited availability of parabolic flights. However, enough data was obtained to reach conclusions about the differences in LOC between normal gravity and microgravity. Without an external radiant flux the LOC of the ETFE wire was 26.4%, which is lower than the LOC in 1g (31.5%). The lowest oxygen concentration at which opposed flame spread was observed was 20.3% for a radiant of 18.8 kW/m2 It is seen that both microgravity and an external radiant flux result in a decrease in the LOC of the ETFE wire. A phenomenological explanation of the results is that microgravity limits flame heat losses, making it possible to reduce flame temperature and maintain the necessary heat transfer to achieve a critical solid decomposition rate. External radiation is able to further compensate for reductions in flame temperature, resulting in lower LOC in 1g and µg. 2. Normal gravity experiments of the flammability of cylindrical polymers exposed to external radiation. An experimental study has been initiated of the burning properties and flammability boundaries of cylindrical polymers with conductive metal cores under external radiation and low velocity flows of varied oxygen concentrations. For the study an experimental set up has been designed and constructed. It is based on a vertically oriented cylindrical sample for testing in normal gravity. The apparatus consists of a flow duct, infrared heaters, and supporting instrumentation. The inlet to the test section is filled with borosilicate beads which act to homogenize the flow, evenly mixing the gases. The test section itself consists of a 254 mm long Pyrex Cylinder with an inner diameter of 70 mm. Below the test section is a flow chamber which allows for controlled flows of air with a prescribed oxygen (balance nitrogen) concentration of between 14 and 30 percent at velocities between 0.2 and 1.0 m/sec. Spaced around the sample are three IR lamps

	capable of irradiating the sample with upwards of 12 kW/m2 of incident flux. The radiant heaters are quartz infrared halogen lamps with a lit length of 127 mm and a total length of 220 mm which are fit into Research Inc. parabolic strip heaters. The three heaters are radially equispaced around the test section at 120 degree intervals, thus irradiating the fuel sample with an approximately uniform heat flux. A hot-wire igniter consisting of an electrically heated Kanthal wire coil in contact with the top surface of the cylindrical sample is used to ignite the fuel sample. The samples are supported at their bottom by an aluminum cylinder. Sample surface temperatures are measured with a type K thermocouple placed halfway along the length of the test specimen. Experiments to date have used High Density Polyethylene (HDPE) samples of three form factors: solid polymer cylinders, hollow polymer cylinders, and polymer cylinders around a core of solid copper. It was found that the burning of the HDPE is characterized by dripping of the molten polymer which brings new issues when comparing with other non-melting polymers like PMMA. These experiments are currently being conducted to determine the LOC boundaries as a function of the external radiant flux. The basic experimental procedure is the same as that used in the experiments described above.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Articles in Peer-reviewed Journals	Osorio AF, Mizutani K, Fernandez-Pello C, Fujita O. "Microgravity flammability limits of ETFE insulated wires exposed to external radiation." Proceedings of the Combustion Institute. 2015;35(3):2683-9. http://dx.doi.org/10.1016/j.proci.2014.09.003 , Jan-2015